

DAY 4: PARTIAL DERIVATIVES AND GRADIENT - NOTES

INTRODUCTION TO MULTIVARIABLE CALCULUS

Goal

Extend calculus to functions of multiple variables. Master partial derivatives and the gradient vector. Understand how these concepts are essential for describing physical quantities that depend on position in space.

1 INTRODUCTION TO MULTIVARIATE FUNCTIONS

In physics, most quantities depend on more than one variable:

- Temperature in a room: $T(x, y, z, t)$ - depends on position and time
- Electric potential: $V(x, y, z)$ - depends on position
- Pressure in a fluid: $P(x, y, z, t)$ - depends on position and time
- Wave function: $\psi(x, y, z, t)$ - depends on position and time
- Gravitational potential: $U(x, y, z) = -\frac{GMm}{\sqrt{x^2+y^2+z^2}}$

1.1 Notation

For a function of two variables: $z = f(x, y)$

- x and y are independent variables
- z is the dependent variable

- The graph is a surface in 3D space

For a function of three variables: $w = f(x, y, z)$

- Cannot be fully visualized (requires 4D)
- Represented by level surfaces: $f(x, y, z) = \text{constant}$
- Example: $x^2 + y^2 + z^2 = R^2$ is a sphere (level surface of $f = x^2 + y^2 + z^2$)

1.2 Visualizing Multivariate Functions

Level Curves (Contour Plots): For $z = f(x, y)$, curves where $f(x, y) = c$ (constant).

- Used in topographic maps (elevation)
- Weather maps (isobars - constant pressure)
- Temperature maps (isotherms)

Level Surfaces: For $w = f(x, y, z)$, surfaces where $f(x, y, z) = c$.

- Equipotential surfaces in electrostatics
- Isothermal surfaces in thermodynamics

2 PARTIAL DERIVATIVES

2.1 Motivation

In single-variable calculus, the derivative $\frac{df}{dx}$ tells us how f changes when x changes. For multivariate functions, we need to know how the function changes when we change one variable at a time, holding the others constant.

2.2 Definition

The partial derivative of $f(x, y)$ with respect to x at (x_0, y_0) is:

$$\frac{\partial f}{\partial x}(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h}$$

Similarly, with respect to y :

$$\frac{\partial f}{\partial y}(x_0, y_0) = \lim_{k \rightarrow 0} \frac{f(x_0, y_0 + k) - f(x_0, y_0)}{k}$$

2.3 Interpretation

- $\frac{\partial f}{\partial x}$ is the rate of change of f as x changes, holding y constant
- $\frac{\partial f}{\partial y}$ is the rate of change of f as y changes, holding x constant
- Geometrically: $\frac{\partial f}{\partial x}$ is the slope of the tangent line in the x -direction on the surface $z = f(x, y)$
- Physically: If $T(x, y)$ is temperature on a plate, $\frac{\partial T}{\partial x}$ tells how temperature changes as you move in the x -direction

2.4 Computation

To compute $\frac{\partial f}{\partial x}$, treat all other variables as constants and differentiate normally with respect to x .

2.4.1 Example 1: Polynomial Function

$$f(x, y) = x^2y + 3xy^2 - 2x + 5y$$

$$\frac{\partial f}{\partial x} = 2xy + 3y^2 - 2$$

$$\frac{\partial f}{\partial y} = x^2 + 6xy + 5$$

2.4.2 Example 2: Trigonometric Function

$$f(x, y) = \sin(x^2y) + e^{xy}$$

$$\frac{\partial f}{\partial x} = \cos(x^2y) \cdot (2xy) + ye^{xy} = 2xy \cos(x^2y) + ye^{xy}$$

$$\frac{\partial f}{\partial y} = \cos(x^2y) \cdot (x^2) + xe^{xy} = x^2 \cos(x^2y) + xe^{xy}$$

2.4.3 Example 3: Rational Function

$$f(x, y) = \frac{x^2+y}{x+y^2}$$

Using quotient rule (treating y as constant for $\partial/\partial x$):

$$\frac{\partial f}{\partial x} = \frac{(2x)(x+y^2) - (x^2+y)(1)}{(x+y^2)^2}$$

2.4.4 Example 4: Function of Three Variables

$$f(x, y, z) = x^2yz^3 + \sin(xz)$$

$$\frac{\partial f}{\partial x} = 2xyz^3 + z \cos(xz)$$

$$\frac{\partial f}{\partial y} = x^2z^3$$

$$\frac{\partial f}{\partial z} = 3x^2yz^2 + x \cos(xz)$$

2.5 Partial Derivatives at a Point

To evaluate a partial derivative at a specific point:

1. Find the general partial derivative
2. Substitute the coordinates

Example: For $f(x, y) = x^2e^{xy}$, find $\frac{\partial f}{\partial x}$ at $(1, 0)$.

$$\frac{\partial f}{\partial x} = 2xe^{xy} + x^2ye^{xy} = e^{xy}(2x + x^2y)$$

$$\text{At } (1, 0): \frac{\partial f}{\partial x} = e^0(2(1) + (1)^2(0)) = 1 \times 2 = 2$$

2.6 Geometric Interpretation

For the surface $z = f(x, y)$:

- $\frac{\partial f}{\partial x}(x_0, y_0)$ is the slope of the curve formed by intersecting the surface with the plane $y = y_0$
- $\frac{\partial f}{\partial y}(x_0, y_0)$ is the slope of the curve formed by intersecting the surface with the plane $x = x_0$

3 HIGHER-ORDER PARTIAL DERIVATIVES

3.1 Notation

Just as with ordinary derivatives, we can take partial derivatives of partial derivatives.

$$\begin{aligned}\frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) && \text{(second partial with respect to } x) \\ \frac{\partial^2 f}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) && \text{(second partial with respect to } y) \\ \frac{\partial^2 f}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) && \text{(mixed partial: } y \text{ then } x) \\ \frac{\partial^2 f}{\partial y \partial x} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) && \text{(mixed partial: } x \text{ then } y)\end{aligned}$$

Alternative notation:

$$f_{xx} = \frac{\partial^2 f}{\partial x^2}, \quad f_{yy} = \frac{\partial^2 f}{\partial y^2}, \quad f_{xy} = \frac{\partial^2 f}{\partial y \partial x}, \quad f_{yx} = \frac{\partial^2 f}{\partial x \partial y}$$

3.2 Example: Computing Higher-Order Partial

For $f(x, y) = x^3y^2 + 2xy^3$:

First partials:

$$f_x = 3x^2y^2 + 2y^3$$

$$f_y = 2x^3y + 6xy^2$$

Second partials:

$$f_{xx} = \frac{\partial}{\partial x}(3x^2y^2 + 2y^3) = 6xy^2$$

$$f_{yy} = \frac{\partial}{\partial y}(2x^3y + 6xy^2) = 2x^3 + 12xy$$

$$f_{xy} = \frac{\partial}{\partial y}(f_x) = \frac{\partial}{\partial y}(3x^2y^2 + 2y^3) = 6x^2y + 6y^2$$

$$f_{yx} = \frac{\partial}{\partial x}(f_y) = \frac{\partial}{\partial x}(2x^3y + 6xy^2) = 6x^2y + 6y^2$$

Notice that $f_{xy} = f_{yx}$!

3.3 Clairaut's Theorem (Equality of Mixed Partial)

If f has continuous second partial derivatives, then:

$$\boxed{\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}}$$

The order of differentiation does not matter for well-behaved functions.

Physical Significance: This is why many physical potentials are "conservative" - the work done is path-independent.

3.4 Example Where Mixed Partial Differ

For most functions encountered in physics, mixed partials are equal. Functions where they differ are pathological and rarely appear in physical applications.

4 THE GRADIENT VECTOR

4.1 The Big Idea: From Slope to Slope Field

In single-variable calculus, the derivative $f'(x)$ tells you the slope of the tangent line at a point — how the function changes as you move along the x -axis.

In multivariable calculus, a function like $f(x, y)$ has a slope in **many** directions. The gradient vector, denoted by ∇f , is the tool that organizes all this information. It is not a single slope, but a **vector** that points in the direction of the steepest ascent and whose magnitude is the slope in that direction.

4.2 Definition

For a function $f(x, y, z)$, the gradient is a vector of all first partial derivatives:

$$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$$

For a function of two variables: $\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$

The symbol ∇ is called "nabla" or "del".

4.3 Examples

Example 1: $f(x, y) = x^2 + y^2$

$$\nabla f = (2x, 2y)$$

Example 2: $f(x, y, z) = xyz$

$$\nabla f = (yz, xz, xy)$$

Example 3: $f(x, y, z) = \frac{1}{\sqrt{x^2+y^2+z^2}}$ (gravitational potential)

$$\frac{\partial f}{\partial x} = -\frac{x}{(x^2 + y^2 + z^2)^{3/2}}$$

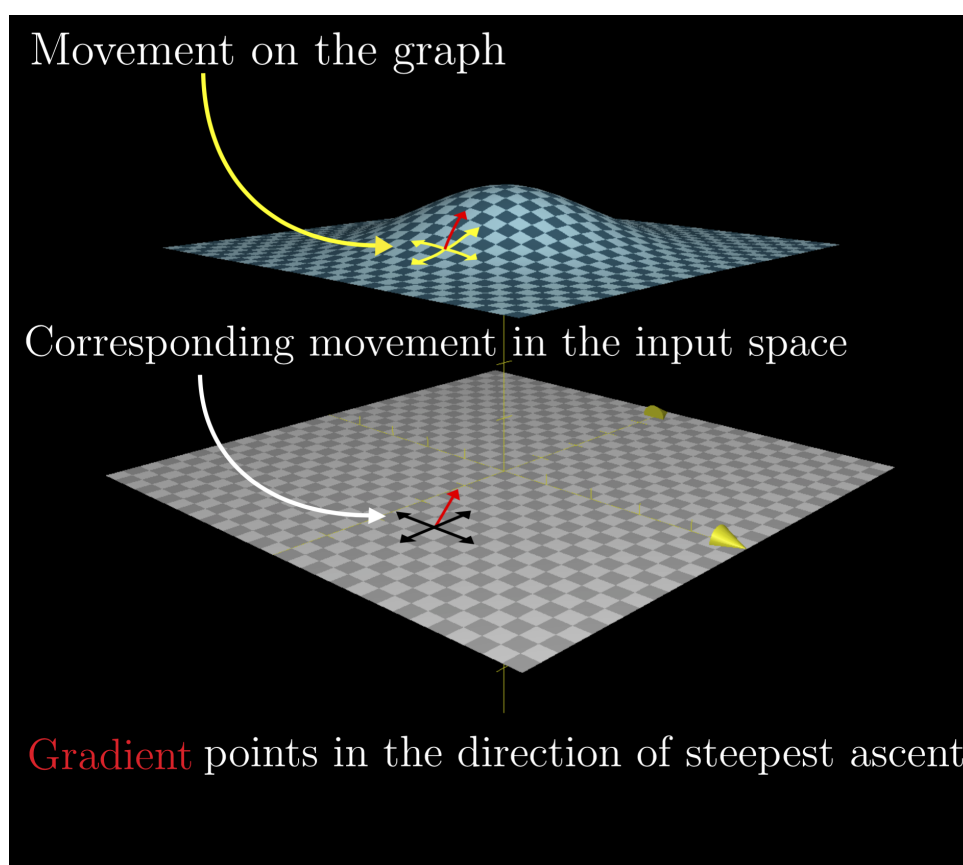
Similarly for y and z , so:

$$\nabla f = -\frac{(x, y, z)}{(x^2 + y^2 + z^2)^{3/2}} = -\frac{\mathbf{r}}{r^3}$$

4.4 Geometric Interpretation

The gradient vector ∇f at a point has two crucial geometric properties:

1. **Direction of Steepest Ascent:** ∇f points in the direction of the greatest increase of f . If you are standing on a hillside described by f , the gradient points directly uphill.
2. **Magnitude = Rate of Ascent:** $|\nabla f|$ gives the rate of increase in that steepest direction. It tells you how steep the hill is in the steepest direction.



4.5 Building Intuition: The Slope in Any Direction

The true power of the gradient is that it allows you to find the slope in **any** direction, not just the steepest one. This is done with the **directional derivative**.

If you have a **unit vector** $\mathbf{u}$ representing a direction of interest, the slope of the function f in that direction is simply the dot product of the gradient with that unit vector:

$$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$$

This formula encapsulates everything:

- If $\mathbf{u}$ points in the same direction as ∇f , the dot product is maximum ($|\nabla f|$). You are going straight uphill.
- If $\mathbf{u}$ points opposite to ∇f , the dot product is minimum ($-|\nabla f|$). You are going straight downhill.
- If $\mathbf{u}$ is perpendicular to ∇f , the dot product is zero. You are moving tangent to the level set, and the function's value isn't changing (instantaneously).

4.6 Properties of the Gradient

1. **Linearity:** $\nabla(cf) = c\nabla f$ for constant c
2. **Sum Rule:** $\nabla(f + g) = \nabla f + \nabla g$
3. **Product Rule:** $\nabla(fg) = f\nabla g + g\nabla f$
4. **Quotient Rule:** $\nabla(f/g) = \frac{g\nabla f - f\nabla g}{g^2}$
5. **Chain Rule:** $\nabla f(g(x)) = f'(g(x))\nabla g(x)$

4.7 Example: Using Gradient Properties

Find $\nabla(r^n)$ where $r = \sqrt{x^2 + y^2 + z^2}$.

We know $\nabla r = \frac{\mathbf{r}}{r}$ from Example 3 above.

Using chain rule: $\nabla(r^n) = nr^{n-1}\nabla r = nr^{n-1}\frac{\mathbf{r}}{r} = nr^{n-2}\mathbf{r}$

5 PHYSICS APPLICATIONS OF THE GRADIENT

5.1 The Gradient as a "Source" of Flow

Think of a physical field, like temperature $T(x, y)$ on a metal plate.

- **Intuitive View:** At any point, the gradient ∇T tells you the direction you'd have to go to warm up your fingers as fast as possible. Its magnitude tells you how quickly the temperature would rise.
- **Visualizing Flows:** The gradient is the fundamental building block for describing flows in physics. It tells you how a quantity is changing in space and thus dictates how other things (like heat or particles) will move in response.

5.2 Conservative Forces and Potential Energy

The most important application of the gradient in physics is the relationship between force and potential energy.

For a conservative force, the force is the **negative gradient** of the potential energy. This means the force points in the direction of the most rapid *decrease* in potential energy:

$$\boxed{\mathbf{F} = -\nabla U}$$

This means:

$$F_x = -\frac{\partial U}{\partial x}, \quad F_y = -\frac{\partial U}{\partial y}, \quad F_z = -\frac{\partial U}{\partial z}$$

A particle in a potential energy landscape will feel a force pushing it from higher to lower potential, in the direction opposite to the gradient.

5.2.1 Example 1: Gravitational Potential Near Earth's Surface

$U = mgz$ (taking z upward)

$$\nabla U = \left(0, 0, \frac{\partial}{\partial z}(mgz)\right) = (0, 0, mg)$$

Therefore:

$$\mathbf{F} = -\nabla U = (0, 0, -mg)$$

This is exactly the gravitational force: downward with magnitude mg .

5.2.2 Example 2: Gravitational Potential of a Point Mass

$U = -\frac{GMm}{r}$ where $r = \sqrt{x^2 + y^2 + z^2}$

From our earlier calculation:

$$\nabla U = GMm \frac{\mathbf{r}}{r^3}$$

Therefore:

$$\mathbf{F} = -\nabla U = -GMm \frac{\mathbf{r}}{r^3}$$

This is the inverse-square law of gravitation: $\mathbf{F} = -\frac{GMm}{r^2} \hat{\mathbf{r}}$

5.2.3 Example 3: Spring Potential

$U = \frac{1}{2}kx^2$ (for a spring along the x-axis)

$$\nabla U = (kx, 0, 0)$$

Therefore:

$$\mathbf{F} = -\nabla U = (-kx, 0, 0)$$

This is Hooke's law: force proportional to displacement, opposite direction.

5.3 Electric Field from Potential

Similarly, in electrostatics, the electric field is the negative gradient of the electric potential:

$$\boxed{\mathbf{E} = -\nabla V}$$

5.3.1 Example: Point Charge Potential

$V = \frac{kQ}{r}$ where $r = \sqrt{x^2 + y^2 + z^2}$

$$\nabla V = -kQ \frac{\mathbf{r}}{r^3}$$

Therefore:

$$\mathbf{E} = -\nabla V = kQ \frac{\mathbf{r}}{r^3}$$

This is Coulomb's law: $\mathbf{E} = \frac{kQ}{r^2} \hat{\mathbf{r}}$

5.4 Heat Flow

Fourier's law of heat conduction states that heat flux $\mathbf{q}$ is proportional to the negative temperature gradient. Heat flows from hot to cold, in the direction of steepest temperature decrease:

$$\mathbf{q} = -k\nabla T$$

5.5 Diffusion

Fick's law of diffusion states that the particle flux $\mathbf{J}$ is proportional to the negative concentration gradient. Particles flow from regions of high concentration to low concentration:

$$\mathbf{J} = -D\nabla C$$

6 SUMMARY OF KEY CONCEPTS

Concept	Formula
Partial derivative w.r.t x	$\frac{\partial f}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h}$
Partial derivative w.r.t y	$\frac{\partial f}{\partial y} = \lim_{k \rightarrow 0} \frac{f(x, y+k) - f(x, y)}{k}$
Second partial	$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right)$
Mixed partial	$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right)$
Clairaut's Theorem	$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$
Gradient (2D)	$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right)$
Gradient (3D)	$\nabla f = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial f}{\partial z} \right)$
Directional derivative	$D_{\mathbf{u}}f = \nabla f \cdot \mathbf{u}$ (where $\mathbf{u}$ is a unit vector)
Conservative force	$\mathbf{F} = -\nabla U$
Electric field	$\mathbf{E} = -\nabla V$
Heat flow	$\mathbf{q} = -k\nabla T$
Diffusion	$\mathbf{J} = -D\nabla C$

Key Physics Relations

Gravitational potential (near Earth): $U = mgz \Rightarrow \mathbf{F} = (0, 0, -mg)$

Gravitational potential (point mass): $U = -\frac{GMm}{r} \Rightarrow \mathbf{F} = -\frac{GMm}{r^2} \hat{\mathbf{r}}$

Spring potential: $U = \frac{1}{2}kx^2 \Rightarrow \mathbf{F} = (-kx, 0, 0)$

Electric potential (point charge): $V = \frac{kQ}{r} \Rightarrow \mathbf{E} = \frac{kQ}{r^2} \hat{\mathbf{r}}$

Heat flow: $\mathbf{q} = -k\nabla T$

Diffusion: $\mathbf{J} = -D\nabla C$